TEXORA · LECTURE SHEET

Crimp of Yarn in Woven and Knitted Fabric

A yarn inside a fabric is never straight: it bends over and under the crossing threads, so it is longer than the fabric it occupies. Crimp measures that extra length, and this sheet covers its geometry, its measurement, and its direct effect on weaving take-up and yarn costing.

TE-3201 · QUALITY EVALUATION OF FABRIC AND APPAREL · TOPIC 2 · LEVEL 3 TERM 2

1. Definition of crimp

When warp and weft interlace, each thread follows a wavy path. Remove a thread from the fabric, straighten it without stretching it, and it is longer than the length of fabric it came from. Crimp expresses that excess as a percentage of the fabric length:

$$\text{Crimp \%} = (\text{straightened yarn length} - \text{fabric length}) / \text{fabric length} \times 100$$

Crimp is defined separately for warp and weft, and the two values in the same fabric are rarely equal. Warp crimp is set by weaving tension, shed geometry and beat-up; weft crimp is set by insertion tension and by how much the warp is forced to bend around the pick. Both change again in finishing: relaxation, dyeing and heat setting allow threads to rearrange, so grey-fabric crimp and finished-fabric crimp are different quantities and must not be mixed in a report.

Crimp must be distinguished from related terms. **Take-up** in weaving is the practical form of warp crimp: the extra warp length the loom consumes to produce a given length of cloth. **Shrinkage** is a change of fabric dimension on washing or relaxation; it is related to crimp because a fabric can shorten by its threads crimping more deeply, but shrinkage is measured on the fabric, crimp on the extracted thread. Confusing the two leads to costing and specification errors.

2. Measurement of crimp

2.1 Marking and extraction

- Condition the fabric in the standard atmosphere, 65% RH and 20 °C, and lay it flat without tension.
- Mark two fine lines square across the thread direction a known distance apart — 25 cm is a convenient laboratory length — taking care that the marks cross the threads to be extracted at right

angles.

- Cut a narrow strip containing the marked length and carefully withdraw one thread at a time from between the marks, avoiding any pull that would stretch the thread or remove its natural set prematurely. Fraying the edges helps each thread to be gripped cleanly.
- Extract at least ten warp threads and ten weft threads from different positions, because crimp varies with position in the weave repeat and across the width.

2.2 Tensioning and reading length

The extracted thread still holds its wave. Its length must be measured straight, but under only enough tension to remove the wave without elongating the yarn itself. In practice the thread is laid along a graduated scale or loaded in a crimp tester, in which a small specified tension — chosen in proportion to the yarn count, just sufficient to straighten it — is applied by a weight or spring and the straightened length is read. Too little tension leaves residual wave and under-reads crimp; too much stretches the yarn and over-reads it. Each thread is read once, decisively: repeated pulling progressively stretches a staple-fibre yarn and the later readings drift upward.

> **WORKED EXAMPLE — WARP AND WEFT CRIMP**
>
> Threads were extracted from a marked fabric length of 25.0 cm.
>
> Warp thread, straightened: 27.2 cm. Warp crimp = (27.2 − 25.0) / 25.0 × 100 = 2.2 / 25.0 × 100 = **8.8%**.
>
> Weft thread, straightened: 26.1 cm. Weft crimp = (26.1 − 25.0) / 25.0 × 100 = 1.1 / 25.0 × 100 = **4.4%**.
>
> The warp is the more crimped system in this fabric — the usual case where warp tension during weaving is released after the cloth leaves the loom and the warp, seeking to recover, bends more deeply around the weft.

Report the mean crimp for each system, the number of threads, the marked length, and the tensioning method. State also whether the fabric was grey or finished, because a result without that history cannot be used for costing a repeat order.

3. Crimp geometry — amplitude, wavelength and spacing

The wavy path of a thread can be described geometrically. Consider one warp thread passing over and under successive picks. The distance between two corresponding points of the wave — for example from the crest over one pick to the crest over the next-but-one pick in a plain weave — is related to the thread spacing, the distance from the centre of one crossing thread to the centre of the next. Three quantities describe the wave:

- **Thread spacing (p)** — the centre-to-centre distance between adjacent crossing threads; its reciprocal is the thread density of the crossing system.
- **Wavelength** — the fabric length occupied by one complete wave of the thread.
- **Crimp amplitude (h)** — the maximum displacement of the thread centre from the centre plane of the fabric; it grows with the diameter of the crossing threads and falls as the thread itself is forced flatter.

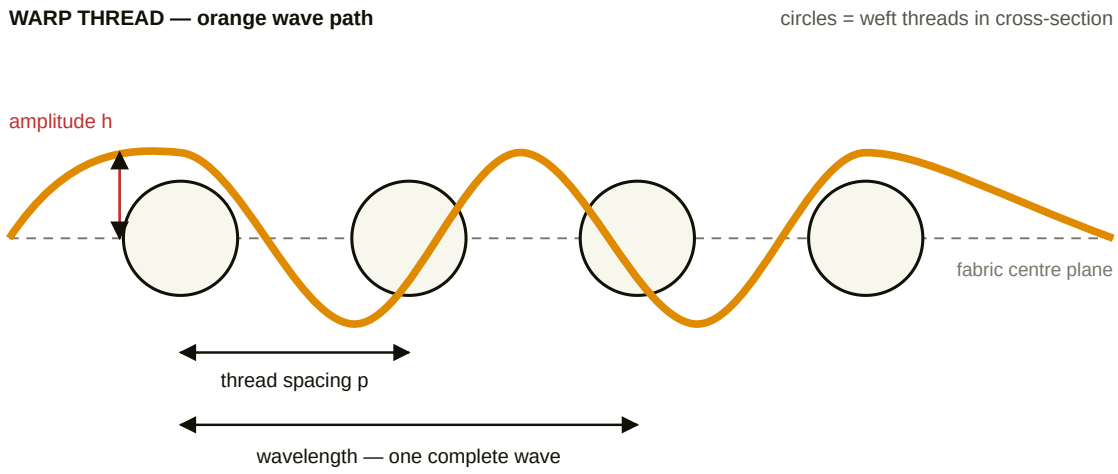


Figure 1 — Crimp geometry. One warp thread waves over and under successive weft threads; spacing, wavelength and amplitude together determine the crimp percentage.

Geometry explains the everyday rules. A coarser crossing yarn forces a larger amplitude, so crimp rises. A closer spacing forces the wave to rise and fall in a shorter distance, so crimp rises again. A stiffer yarn resists bending and pushes more of the wave into the crossing system instead. In the limiting models of fabric geometry the thread path is treated first as straight inclined segments and then as a curve, but for laboratory work the message is the measured one: crimp rises with crossing-thread diameter and density, and falls when the thread can flatten at the interlacing.

4. Crimp interchange and balance in fabric design

Warp and weft compete for the total bending that interlacing requires. If the warp is held very taut on the loom, it stays comparatively straight and the weft must do most of the bending: warp crimp is low, weft crimp is high. Release that tension, or weave with a slack warp, and the balance reverses. This exchange is called crimp interchange, and a designer uses it deliberately:

- **Balanced crimp** — similar warp and weft crimp gives a fabric that behaves alike in both directions, shrinks evenly and presents an even surface; a natural target for plain apparel fabrics.
- **Warp-dominant surface** — low warp crimp with the warp held straight and prominent is used where warp threads must cover the surface, as in warp-faced constructions; the weft hides by

crimping deeply.

- **Effects on properties** — the more crimped system extends more before its threads straighten and begin to carry load, so it shows the higher initial extension and the later rise of tensile load; it also shrinks more on relaxation because it has more length stored in its wave.

Crimp balance is therefore checked whenever a fabric tears unexpectedly in one direction, shrinks unevenly, or shows a ribbed surface that the weave alone does not explain. The fault is often not the weave but a tension change at the loom that silently moved crimp from one system to the other.

5. Crimp, take-up and yarn-consumption costing

Crimp is money. The loom consumes yarn at the straightened length, not at the fabric length, so every costing that multiplies fabric length by ends or picks without a crimp allowance under-orders yarn by exactly the crimp percentage. Warp take-up is the mill expression of warp crimp:

$$\text{Warp length required per end} = \text{fabric length} \times (1 + \text{crimp \%} / 100)$$

WORKED EXAMPLE — WARP LENGTH FOR 100 M OF FABRIC

Warp crimp from the worked example above is 8.8%.

Warp yarn needed per 100 m of fabric = 100 × (1 + 8.8/100) = 100 × 1.088 = **108.8 m per end**.

For a warp of, say, 4,000 ends this is 435,200 m of warp yarn for 100 m of cloth before any waste allowance — a mill that costed on 400,000 m would be short by 35,200 m and would discover the error at the loom, not at the desk.

The same correction applies to weft: pick length must cover fabric width plus crimp plus selvedge waste. In fabric weight calculation, crimp enters both systems: the mass of warp per square metre is proportional to ends per unit width × yarn tex × (1 + warp crimp), and likewise for weft. A change of finish that raises crimp by two percentage points raises yarn consumption and fabric mass together, even though no thread count has changed — a frequent hidden cause of a finished fabric arriving heavier than its grey specification.

6. Crimp in knitted fabrics

A knitted loop is crimp in a different geometry. The yarn in a loop is bent in three dimensions: around the needle loop, through the sinker loop, and into the intermeshing points with the course below. There is no straight fabric length to mark and extract in the woven sense; instead the corresponding quantity is loop length — the length of yarn in one stitch — measured by unravelling a counted number of courses, exactly as in the woven extraction method, and dividing by the number of loops unravelled.

The ideas transfer directly. A longer loop length at fixed course and wale counts means more yarn stored in the structure: the knit is looser, more extensible, heavier per unit area than its counts alone suggest, and more prone to width growth and length shrinkage on relaxation. A short loop length gives a tight, firm, more stable fabric. Just as woven crimp interchange moves bending between warp and weft, loop shape in a knit distributes yarn between the width and length directions of the loop, which is why knitted dimensional stability is always discussed through loop length and loop shape together, never through counts alone.

Glossary

Crimp % — excess length of a straightened thread over the fabric length it occupied, as a percentage of the fabric length. **Take-up** — warp crimp expressed as the extra warp length consumed in weaving. **Crimp amplitude** — maximum displacement of a thread centre from the fabric centre plane. **Wavelength** — fabric length of one complete wave of a thread. **Thread spacing** — centre-to-centre distance between adjacent crossing threads. **Crimp interchange** — transfer of bending between warp and weft when tensions change. **Loop length** — length of yarn in one knitted stitch. **Crimp tester** — instrument that straightens an extracted thread under a small specified tension for length reading.

Key points

• Crimp % = (straightened length − fabric length) / fabric length × 100, measured separately for warp and weft.

• Extract threads from a marked length and straighten under just enough tension to remove the wave without stretching the yarn.

• Crimp rises with crossing-thread diameter and density; amplitude, wavelength and spacing describe the wave.

• Warp and weft crimp interchange with weaving tension — balance is a design choice with direct property consequences.

• Yarn consumption = fabric dimension × (1 + crimp/100) in each system; ignoring crimp under-orders yarn by the crimp percentage.

• In knits the corresponding quantity is loop length, measured by unravelling a counted number of loops.

Practice

1. A warp thread extracted from 20.0 cm of fabric straightens to 21.5 cm, and a weft thread from the same length straightens to 20.7 cm. Calculate both crimp percentages and state which system is more crimped.

2. Weft crimp is 5.5%. Calculate the length of weft yarn in one pick across a fabric 160 cm wide, ignoring selvedge waste, and the total weft yarn length for 1,000 picks.
3. A mill costs warp yarn as exactly 100 m per end for 100 m of cloth, but warp crimp is 9.5%. Calculate the actual requirement per end and the shortage over 3,600 ends.
4. Explain, using crimp interchange, why raising warp tension on the loom can increase weft crimp and change the fabric surface, even though the weave and thread counts are unchanged.

Answer key

1. Warp crimp = (21.5 − 20.0) / 20.0 × 100 = 1.5 / 20.0 × 100 = **7.5%**. Weft crimp = (20.7 − 20.0) / 20.0 × 100 = 0.7 / 20.0 × 100 = **3.5%**. The warp is more crimped, by 4.0 percentage points.
2. Pick length = 160 × (1 + 5.5/100) = 160 × 1.055 = **168.8 cm**. For 1,000 picks: 168.8 × 1,000 = 168,800 cm = **1,688 m**.
3. Actual requirement = 100 × 1.095 = **109.5 m per end**. Shortage per end = 9.5 m. Over 3,600 ends: 9.5 × 3,600 = **34,200 m** short.
4. Interlacing requires a fixed total amount of bending shared between the two systems. A taut warp resists bending and stays straighter, so the weft must bend more deeply around it at every interlacing: weft crimp rises while warp crimp falls. The straighter, more prominent warp also covers more of the surface, so the fabric looks more warp-dominant — a change of handle, extension and appearance produced by tension alone.

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